

GHANA COMMUNICATION TECHNOLOGY UNIVERSITY



FACULTY OF ENGINEERING
DEPARTMENT OF COMPUTER ENGINEERING

DEVELOPMENT AND IMPLEMENTATION OF A ROBOTIC APPLICATION FOR HUMAN INTENTION RECOGNITION USING MOTION AND HAND GESTURE

*A Final Project Report Submitted in Partial Fulfillment of the Requirements for the
Degree of Bachelor of Science in Computer Engineering*

PRESENTATION BY:

ELEANA OSEI OWUSU — 4121230024

JOEL NII ADJETEY AHULU — 4121230020

SUPERVISOR: MR. MICHEAL XENYA

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PRESENTATION OUTLINE

- 01. Introduction, Problem Statement & Objectives**
- 02. Literature Review & Critical Research Gaps**
- 03. System Architecture & 6-Phase Methodology**
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- 05. Experimental Results & ISO Safety Compliance**
- 06. Conclusion & Physical Demonstration Transition**



01. INTRODUCTION & RESEARCH BACKGROUND

Robots Working Directly with Humans

Robots are leaving safety cages to work alongside people in hospitals, warehouses, and homes. For humans and robots to coexist safely, the robot must understand where people are and respect their personal space.

Natural, Hands-Free Interaction

Instead of requiring handheld controllers, touchscreens, or keyboards, humans can guide the robot using natural hand gestures and body movement—just like signaling to a human coworker.

Fast Onboard Intelligence (No Cloud Lag)

Sending live video over the internet causes dangerous delays (>200 ms) and fails when Wi-Fi drops. In this project, all AI recognition and navigation decisions happen entirely inside the robot in real time.



Figure 1.1: Collaborative Human-Robot Interaction (HRI) in an industrial shared workspace, illustrating contactless gesture coordination.

02. PROBLEM STATEMENT

Current mobile robots separate vision recognition from active navigation, relying on remote cloud servers that suffer from high network lag (>200 ms) and connection dropouts.

In shared human-robot workspaces, this delay prevents robots from reacting in real time to human gestures, creating severe collision hazards under ISO 15066 safety standards.

PERCEPTION-ACTION GAP

Vision AI isolated from wheel motors & obstacle avoidance
(Tsitos et al., 2022)

CLOUD LATENCY HAZARD

Network lag > 200 ms & Wi-Fi drops
violate ISO 15066 safe stopping
margins

EDGE COMPUTING LIMIT

Heavy vision AI overheats mobile chips; demands a sub-135 ms
onboard pipeline



03. PROJECT OBJECTIVES

GENERAL OBJECTIVE

To design, implement, and evaluate an edge-deployed, real-time human intention recognition and autonomous navigation system powered by ROS 2 on an embedded mobile robot.

01. Invariant Vision Pipeline

Compute 19 scale-invariant geometric features from 21 hand landmarks for real-time onboard MLP classification.

02. Balanced 6,000-Sample Dataset

Capture and balance a custom 6-class dataset directly through the onboard camera to eliminate perspective domain shift.

03. LSTM Trajectory Prediction

Forecast short-horizon operator motion trajectories from continuous body pose landmarks using recurrent neural networks.

04. Supervisory Brain Node

Execute a deterministic finite state machine with 5-frame rolling consensus, active operator lock, and visual servoing.

05. 2D LiDAR SLAM & Nav2

Deploy real-time Cartographer mapping and the Nav2 navigation stack for socially aware, collision-free path execution.

06. Empirical Safety Benchmarking

Quantify end-to-end latency (132 ms), physical classification accuracy (96.7%), and compliance with ISO 15066 safety margins.



04. LITERATURE REVIEW & RESEARCH GAPS

CRITICAL APPRAISAL OF CORE BENCHMARK STUDIES (THESIS CHAPTER 2)

Athanasios Tsitos, Maria Dagioglou & Theodoros Giannakopoulos (2022)

Competitive Reaching Intention Game

- **Objective:** Predict human reaching intent early to govern robot arm trajectories within a 150 ms reaction threshold.
- **Methodology:** RGB-D camera + OpenPose wrist tracking + SVM on a 6-DOF UR3 industrial robotic arm.
- **Research Gap:** Confined to a stationary desk — lacks mobile base movement and provides no explicit hand gesture command channel.
- **Project Resolution:** Direct vision-to-wheels coupling for mobile navigation guided by 6 touchless hand gestures.

Jubayer Al Mahmud (2022)

3D Gesture Recognition & Adaptation

- **Objective:** Classify 3D pointing and dynamic hand gestures in real time to guide robot interaction across age groups.
- **Methodology:** Kinect v2 depth sensor + 3D skeletal normalization + CNN / SVM classifiers on 3,600 samples.
- **Research Gap:** Relies on heavy cloud / GPU computers; vulnerable to network latency (>200 ms) and Wi-Fi connection drops.
- **Project Resolution:** Executes 100% of vision AI directly onboard a low-cost Raspberry Pi 5 with zero cloud dependency.

Y. Li, H. Zhang, Guang Yang & Shuoyu Wang (2023)

Intention-Aware Motion Planning

- **Objective:** Plan mobile robot paths around moving site workers by predicting whether obstacles will clear the hallway.
- **Methodology:** 2D LiDAR + camera CNN object detection + costmap updates in Isaac Sim virtual simulation.
- **Research Gap:** Tested only in computer simulation — never validated on physical robot hardware; lacks touchless gesture steering.
- **Project Resolution:** Physically validated on a real mobile robot with 2D LiDAR SLAM, active visual servoing, and ISO 15066 safety stops.

